

REVIEW

Open Access



Exploring the ethical landscape of AI in academic writing

Navneet Ateriya^{1*}, Nagendra Singh Sonwani², Kishor Singh Thakur³, Arvind Kumar⁴ and Satish Kumar Verma⁵

Abstract

Background The incorporation of artificial intelligence (AI) into the process of writing research papers has introduced a new era characterized by enhanced efficiency and production in the field of academia. Nevertheless, this development raises numerous ethical concerns that require in-depth debate.

Main body The present study reviews available literature to investigate the ethical implications of utilizing AI in the process of preparing manuscripts including data analysis, drafting, and publication. Key ethical implications are issues surrounding intellectual property rights, correct attribution in collaborative writing endeavors, and transparency and responsibility for the utilization of AI. The article also examines the implications of inequality in academic publishing and its broader impact on society, including the exacerbation of disparities in access to scientific information.

To tackle these difficulties, the study proposes the establishment of comprehensive rules and optimal methods to guarantee the ethical utilization of AI in the process of writing research papers. Researchers can effectively utilize AI technology while minimizing ethical hazards by promoting transparency, accountability, and diversity.

Conclusion This study offers a comprehensive examination of the ethical factors associated with writing research papers using artificial intelligence. It draws upon a wide range of literature from several disciplines, including ethics, AI, and scholarly publications, and provides in-depth knowledge for scholars, policymakers, and practitioners alike.

Key points

- Ethical AI utilization—AI should support research without compromising academic integrity.
- Academic integrity protection—prevent AI-induced plagiarism through strict citation and verification protocols.
- Transparency in AI usage—researchers must disclose AI assistance in their work to maintain credibility.
- Data privacy and security—ethical AI use requires safeguarding sensitive research data.
- Human oversight in AI applications—AI should aid but not replace human judgment in academic work.
- AI in peer review—AI can enhance reviewer selection but must be monitored for fairness and transparency.

Keywords Artificial intelligence, Academic writing, Authorship attribution, Plagiarism detection, Bias mitigation

*Correspondence:

Navneet Ateriya
drnavneeta@aiimsgorakhpur.edu.in

Full list of author information is available at the end of the article

Background

Artificial intelligence (AI) is a term applied to a machine or software and refers to its capability to simulate intelligent human behavior, instantaneous calculations, problem-solving, and evaluation of new data based on previously assessed data (Amisha, 2019; Causio, 2024). The integration of AI into several academic fields has brought about significant transformations in the generation, examination, and distribution of information in recent times. One of the most intriguing academic applications of AI is its capacity to assist scholars in writing research papers. AI-powered solutions streamline the research process and enhance productivity by offering a wide array of capabilities, including data analysis, literature review, drafting, and editing. However, as AI becomes increasingly incorporated into academics, it gives rise to a multitude of ethical concerns that necessitate meticulous deliberation (Mondal & Mondal 2023; Chetwynd 2024; Ong 2024).

The utilization of AI in research paper authoring raises significant ethical concerns, mostly centered around transparency and accountability. The issue of how much information academics should disclose on the involvement of AI in paper writing is brought up due to the autonomous content generation capabilities of AI algorithms. Moreover, the utilization of AI systems in collaborative writing gives rise to concerns regarding intellectual property rights and the accurate attribution of contributions (Mittelstadt, 2016; Ong, 2024).

The potential existence of partiality and deceptive data in AI systems presents a significant ethical quandary. AI technologies, which are powered by artificial intelligence, may inadvertently maintain and perpetuate existing biases present in the data used for training. This can lead to biased study findings and the reinforcement of systemic inequities, as noted by (Wallach, 2010; Stenseke 2024). Furthermore, the application of AI in generating persuasive narratives gives rise to ethical considerations surrounding the authenticity and integrity of research results. The ethical quandaries concerning AI in research go beyond the immediate act of writing and have significant societal consequences (Inglada, 2024). The increasing reliance on AI in research can sustain disparities in the availability of scientific information, therefore exacerbating the divide between institutions with abundant resources and those with limited ones.

This literature review utilizes a variety of multidisciplinary sources, including scholarly publications, artificial intelligence, and ethics, to deliver a thorough overview of the ethical concerns related to the use of AI in creating research articles. This paper seeks to clarify the principal issues and discussions around AI-generated academic literature by synthesizing insights from many disciplines.

Applications of AI in academic writing

GPT in academic writing

Academic writing tasks have utilized powerful language models such as GPT-3 and GPT-4 developed by OpenAI. Challenges related to authorship and intellectual property have emerged, despite the significant potential of the subject. To guarantee transparency and proper attribution, researchers using GPT must navigate these ethical considerations diligently (Bryson & Theodorou 2019). When GPT significantly contributes to a research article, the human authors should acknowledge the extent of GPT's involvement and ensure that the AI's output is meticulously validated and vetted.

Utilizing GPT in academic writing reveals the ethical quandaries related to AI authorship and intellectual property. While GPT can generate content that is coherent and relevant to the given context, researchers must approach these ethical concerns with caution and diligence. This involves ensuring that there is transparency and proper attribution, especially in situations where GPT makes a significant contribution to a scientific article. Human authors must exercise oversight over the output of the AI, carefully reviewing and validating the material to ensure its truthfulness and compliance with ethical standards (Park 2024).

AI in peer review

AI has been utilized to facilitate the process of peer review, which has prompted issues around accountability and transparency. Ensuring that AI systems do not introduce bias and maintain the integrity of the peer review process is of utmost importance. AI can expedite the review process by finding potential reviewers and detecting instances of plagiarism. Nevertheless, human oversight is still necessary to ensure that the ultimate selections are unbiased and equitable.

Utilizing AI in the peer review process presents both benefits and drawbacks. AI can accelerate the review process by identifying potential reviewers and detecting plagiarism. Nevertheless, it also raises concerns over accountability and openness. Ensuring the integrity of the peer review process and preventing bias is of utmost importance when it comes to AI systems. To ensure that the final results are fair and objective, and to maintain openness and accountability in the AI's participation during the review process, it is necessary to have human oversight (Mittelstadt, 2016; Ong, 2024).

AI-generated research proposals

AI has been employed on a global scale to produce research recommendations. This technique has sparked discussions over academic integrity and the authenticity

of the work. It is crucial to establish explicit norms and ethical standards to effectively address these challenges (Kacena, 2024). AI-generated proposals should be utilized as an initial framework, but they require substantial human involvement and adjustment to guarantee originality and compliance with ethical guidelines.

Utilizing AI for the creation of research proposals in India has ignited discussions around academic integrity and the authenticity of the work. To address these problems, it is crucial to create explicit norms and ethical standards when utilizing AI to draft research proposals. AI-generated proposals should be utilized as an initial framework, but it is crucial to incorporate substantial human involvement and make revisions to guarantee uniqueness and compliance with ethical guidelines. This methodology guarantees that the contributions made by AI adhere to ethical standards and that the ultimate recommendations uphold academic integrity (Hovy & Spruit 2016; Grieve, 2025).

AI in scientific discovery

AI has also been utilized in scientific study, particularly in the domains of material science and the advancement of medicine. While AI can accelerate the discovery process, it is crucial to ensure that the data used is obtained ethically and that the results are validated through rigorous testing (Cath, 2018). In addition, researchers must maintain transparency and integrity on the role of AI in their findings, as well as acknowledge the limitations and inherent biases of AI systems.

The integration of AI in scientific fields such as material science and medicine development has the potential to expedite research and development processes. However, it is crucial to ensure that the information employed in these techniques originates from ethical sources and that the outcomes are validated by meticulous testing. Researchers must recognize the limitations and potential biases of AI systems, as well as the impact that AI plays in their discoveries. This approach ensures the ethical validity of AI's contributions to science and the academic honesty of the resulting research (King, 2017).

To maintain the authenticity of academic research, it is crucial to address the ethical concerns associated with employing AI for the composition of research publications. To harness the advantages of AI in academia, it is important to establish clear guidelines on authorship, intellectual property, data privacy, academic integrity, and bias, thereby ensuring ethical standards are upheld. Future investigations should focus on developing comprehensive frameworks and protocols that govern the ethical utilization of AI in academic writing. Furthermore, to adapt to the swiftly evolving realm of artificial

intelligence, politicians, ethicists, and researchers must have ongoing communication. (Nemitz 2018).

Authorship and intellectual property

Authorship attribution

Authorship attribution is a prominent ethical concern. Traditional notions of authorship are based on a significant intellectual contribution, but AI confuses this need. There are apprehensions over the appropriate attribution of credit to AI systems for their work and whether AI can be considered as an author. Authorship attribution must consider the degree of human agency and the level of AI involvement. To maintain academic honesty, it is necessary to establish certain guidelines that can distinguish between writing that has been aided by AI and writing that has been authored by a human (Chetwynd 2024; Ong, 2024).

Within the realm of AI, the concept of authorship becomes increasingly complex and nuanced. The basis of conventional academic authorship is in the intellectual contributions of scholars. However, when utilizing AI systems like GPT, the input from the human author is combined with the AI systems. Even if an AI generates substantial sections of a research publication, the human researcher remains responsible for the content, assuring its accuracy, coherence, and adherence to ethical norms (Heidt 2024).

Intellectual property rights

The utilization of AI to create content raises concerns regarding intellectual property rights. Identifying the authorship of writing generated by an AI system might provide challenges. Typically, the individual responsible for designing the AI or the user who provided instructions to it will possess the intellectual property (Yousaf, 2025). However, it is crucial to thoroughly contemplate the ethical implications of possessing such ownership, especially when taking into account the initial creative input. The legal frameworks concerning intellectual property rights are still evolving to tackle the specific challenges posed by AI-generated content (Inglada, 2024).

Within the scope of AI, the process of ascertaining intellectual property rights encompasses various entities, such as the developers of the AI, its users, and perhaps the individuals whose data was utilized to train the AI. Given the complex network of interests involved, it is imperative to reassess existing intellectual property policies to ensure they can effectively address the intricacies introduced by AI technology. To prevent conflicts and ensure accurate acknowledgment, it is necessary to establish specific standards for

determining the ownership and usage rights of information produced by AI (Jairoun, 2024).

Data privacy and confidentiality

Handling sensitive data

For AI systems to function well, they require vast quantities of data, a significant portion of which may include private or sensitive information. Safeguarding the privacy and confidentiality of data is of utmost importance. Compliance with data protection regulations such as GDPR and HIPAA is essential when employing ethical AI. This ensures that all data is properly anonymized and securely stored (Mittelstadt, 2016; George 2024; Ong, 2024). To ensure the protection of data from unauthorized access and breaches, researchers must take further measures to encrypt it during both transmission and storage, as well as anonymize it.

Data privacy is an essential subject in AI research due to the increasing amount of data that AI systems are collecting and evaluating. Strict compliance with data protection legislation and processes is crucial for the ethical utilization of data in artificial intelligence. Researchers must ensure the secure transmission, anonymization, and storage of the data utilized by AI systems. To protect sensitive data from breaches and unauthorized access, it is necessary to implement robust encryption mechanisms and access control systems (O'Neil 2016).

Transparency and consent

Obtaining informed consent and ensuring transparency regarding the data utilized by AI systems is crucial. Researchers must obtain participants' explicit consent and ensure their comprehension of how their data will be utilized, including any potential AI analysis (Noble 2018). This involves providing a comprehensive description of the role of AI in research, the types of data that are collected, how they will be utilized, and any potential risks. Transparency promotes confidence and assists in maintaining ethical standards in research protocols.

Within the realm of AI research, informed consent refers to the act of providing participants with comprehensive details regarding the utilization of their data. Researchers must ensure that participants have a clear comprehension of how their data will be utilized, including any potential analysis using AI. This necessitates transparency regarding the use of AI in research, the kind of data collected, and any potential apprehensions. To maintain ethical standards and cultivate trust in AI research, it is imperative to obtain explicit agreement from participants (Brown, 2020; Daungsupawong, 2024).

Improper use of AI tools

AI techniques have the potential to be misused to create fraudulent data or generate counterfeit research articles. To maintain academic integrity, it is necessary to set clear ethical guidelines and ethically employ AI techniques (Hosseini, 2024). Publishers and academic institutions must establish and enforce norms that explicitly forbid the inappropriate utilization of AI, including rigorous peer review protocols and mechanisms for detecting and deterring fraudulent activities.

The misuse of AI tools to falsify data or produce fraudulent research articles poses a significant danger to academic integrity. To address this issue, institutions and publishers should establish clear and explicit ethical guidelines and regulations that explicitly forbid the inappropriate utilization of AI. This entails implementing rigorous protocols for peer review and authentication to detect and deter fraudulent behavior. The maintenance of ethical standards and the preservation of the integrity of academic research rely on guaranteeing the conscientious utilization of AI tools (Sandvig, 2014).

Academic integrity and plagiarism

The use of AI technology significantly increases the risk of unintentional plagiarism as it can generate content that closely resembles previously published works. To mitigate this risk, researchers must ensure that they adhere to proper citation protocols and utilize plagiarism detection tools (Tennant 2018). In addition, it is imperative to establish protocols for assessing AI-generated content to ensure originality, acknowledge proper attribution, and avoid the potential hazards of inadvertent plagiarism.

Occasionally, AI-generated content closely imitates pre-existing works, potentially leading to unintentional plagiarism. The utilization of extensive datasets, often containing copyrighted or previously published material, significantly amplifies this risk in AI systems. To mitigate this risk, researchers must utilize plagiarism detection technologies and adhere to proper citation protocols. To mitigate the risks of inadvertently plagiarizing previous material, it is advisable to establish specific standards for assessing AI-generated content. This will help ensure its originality and accurate attribution (Mondal & Mondal 2023).

Causes of plagiarism in academic writing

Several factors contribute to plagiarism in academic settings:

- a) Lack of awareness: some individuals may not fully understand what constitutes plagiarism or the ethical standards of academic writing.
- b) Poor writing skills: inadequate proficiency in writing can lead individuals to copy existing work instead of articulating their ideas.
- c) Pressure to publish: the “publish or perish” culture in academia can drive researchers to engage in unethical practices to meet publication demands.
- d) Unfamiliarity with citation practices: not knowing how to properly cite sources can result in unintentional plagiarism.

These factors highlight the need for comprehensive education on research ethics and academic writing standards (Kwon, 2024; Kampa, 2025).

Remedies for plagiarism

To address plagiarism effectively following remedies can be employed:

1. Education on research ethics: institutions should provide training on ethical writing practices and the importance of originality in research.
2. Development of writing skills: workshops and courses can help individuals improve their writing abilities, reducing reliance on copying existing work.
3. Use of plagiarism detection software: implementing tools to detect similarities can deter plagiarism and promote originality.
4. Clear institutional policies: establishing and enforcing policies with defined consequences for plagiarism can serve as a deterrent.

These strategies aim to foster a culture of integrity and respect for intellectual property in academic environments (Juyal, 2015; Hilliard, 2023).

Bias and fairness

Algorithmic bias

AI systems have the potential to strengthen and magnify existing biases in training data. Researchers must ensure that the data used for training AI systems is unbiased and inclusive. This issue can be mitigated through regular algorithm updates and checks. Moreover, it is essential to incorporate diverse viewpoints in the creation and implementation of AI systems to guarantee fairness and mitigate the potential for reinforcing current disparities. Algorithmic bias is a significant subject of study in the field of AI due to the potential for AI systems to amplify existing prejudices present in their training data. To overcome this issue, researchers must ensure that the data utilized to train AI systems is both unbiased

and representative. To detect and mitigate any biases, it is necessary to conduct regular audits and make adjustments to the algorithms. In order to ensure fairness and reduce the risk of worsening existing disparities, it is crucial to include a wide range of perspectives when developing and implementing AI systems (Bender, 2021; Daungsupawong, 2024).

Fair representation

Ensuring fair and equal representation in AI-generated research is of utmost importance. Researchers must use vigilance to ensure equitable and unbiased outcomes while remaining vigilant for any potential for AI to manipulate results or offer prejudiced interpretations. This involves critically evaluating the outputs of AI systems, including feedback from many groups, and continuously improving the algorithms to promote fairness and inclusivity (Greene, 2019).

In order to guarantee equitable representation in AI-generated studies, it is imperative to maintain the integrity of academic research. Researchers must use vigilance to ensure equitable and unbiased outcomes while remaining vigilant for any potential for AI to manipulate results or present prejudiced interpretations. This involves critically evaluating the outputs of AI systems, considering input from various groups, and continuously improving the algorithms to promote fairness and inclusivity. Ensuring equal participation in AI research is crucial for maintaining moral standards and improving social justice (Tang, 2024).

Causes of bias in AI systems

Bias in AI systems can arise from various sources:

- a) Non-diverse training data: if the data used to train AI models lacks diversity, the AI may not perform well across different populations.
- b) Historical biases: AI systems trained on data reflecting existing societal biases can perpetuate these biases in their outputs.
- c) Algorithm design: choices made by developers during the creation of AI algorithms can inadvertently introduce bias.
- d) Lack of representation: underrepresentation of certain groups in training data can lead to biased AI behavior against those groups.

Recognizing these causes is the first step toward developing fair and unbiased AI systems (Nazer, 2023; Cross, 2024).

Remedies for bias in AI systems

To mitigate bias in AI:

- a) Diversify training data: ensure that datasets used for training AI models are representative of all relevant populations.
- b) Regular audits: conduct ongoing evaluations of AI systems to identify and address potential biases.
- c) Inclusive development teams: involving individuals from diverse backgrounds in the development process can help identify and mitigate biases.
- d) Transparency and accountability: developers should document their design choices and be open about the limitations of their AI systems.

Plagiarism and bias are significant ethical concerns in academic writing and the application of artificial intelligence (AI). Understanding the underlying causes of these issues and implementing effective remedies is crucial for maintaining integrity and fairness in research. Implementing various measures can lead to more equitable AI applications and reduce the risk of perpetuating existing biases (Mittermaier, 2023; Nazer, 2023).

Case studies on AI-induced plagiarism and bias

The integration of artificial intelligence (AI) in academic writing has introduced both opportunities and challenges, particularly concerning issues of plagiarism and bias. Several case studies have highlighted these ethical dilemmas, underscoring the need for robust guidelines and vigilant oversight.

Case studies on AI-induced plagiarism

Increased AI usage in academic submissions The use of large language models (LLMs) such as ChatGPT 3.5 in article submissions has significantly increased since its release, according to a study published in the *Orthopaedics & Traumatology: Surgery & Research* journal. Concerns over originality and possible plagiarism in scientific research were raised by the study's findings that 41% of the publications had a suspected AI usage rate higher than 10% (Maroteau, 2023).

Challenges in detecting AI-generated content A cross-disciplinary systematic review highlighted that while AI tools such as ChatGPT assist in enhancing grammar and coherence in academic writing, they also pose ethical issues, including the risk of plagiarism. The study emphasized the necessity for improved detection mechanisms to identify AI-generated content in scholarly work (Fatima, 2024).

Plagiarism detection in AI-generated research A recent survey focused on plagiarism detection and looked at how LLMs like ChatGPT and Gemini affect academic

integrity. The results showed that artificial intelligence (AI)-generated content frequently evades conventional plagiarism detectors, requiring the creation of sophisticated detection tools to maintain academic standards (Pudasaini, 2024).

Case studies on AI-induced bias

Bias in wildlife research using AI A study on large-scale wildlife research utilizing AI technologies found that bias-prone indices accounted for approximately 50% of outputs. This consistency across user groups highlights the potential for AI systems to perpetuate existing biases, emphasizing the need for critical evaluation and mitigation strategies in research methodologies (Bruce, 2025).

AI bias in medical diagnostics Studies on the use of AI technologies for the diagnosis of temporomandibular disorders evaluated the diagnostic precision of AI models and methodically compiled the body of available material. The study emphasized how crucial it is to overcome any biases in AI algorithms in order to provide fair and precise medical diagnosis (Jha, 2022).

Language bias in systematic reviews An investigation into the exclusion of non-English publications from systematic reviews revealed minimal effects on overall conclusions. However, it also highlighted the potential for language bias, suggesting that AI tools used in evidence synthesis should account for linguistic diversity to avoid skewed outcomes (Nussbaumer-Streit, 2020).

These case studies illustrate the multifaceted ethical challenges posed by AI in academic writing, particularly concerning plagiarism and bias. They underscore the imperative for developing comprehensive ethical guidelines, enhancing detection mechanisms, and fostering critical awareness among researchers to ensure the integrity and fairness of scholarly work.

AI in peer review: enhancing efficiency, reviewer selection, and transparency

AI is increasingly being integrated into the peer review process, offering potential enhancements in efficiency, reviewer selection, and transparency. However, its adoption raises concerns regarding accountability, bias, and the preservation of the review process's integrity.

Reviewer selection

AI systems can streamline the reviewer selection process by analyzing large datasets to match manuscripts with suitable reviewers based on expertise, past performance,

and potential conflicts of interest. For instance, in China, AI has been employed to select grant reviewers, aiming to reduce biases and improve the efficiency of the selection process. However, the effectiveness and fairness of such AI-driven selections require careful evaluation to ensure they meet the desired standards of impartiality and competence (Parrilla, 2023; Krishnan 2024).

Review transparency

The integration of AI in peer review also brings forth challenges related to transparency. A scoping review identified issues in datasets used to develop and test clinical AI algorithms, highlighting a lack of transparency and potential biases that could affect the reliability of AI-assisted reviews.

Furthermore, the absence of established editorial policies for AI tool usage in peer review reports raises ethical concerns regarding transparency and accountability. Addressing these issues is crucial to maintaining trust in the peer review process (Daneshjou, 2021; Garcia 2024).

Human oversight and ethical considerations

Despite AI's potential to expedite reviewer selection and enhance transparency, human oversight remains essential. Over-reliance on AI can introduce biases inherent in algorithms, and ethical concerns related to data privacy and accountability persist. A narrative review emphasized the need for balancing AI's benefits with potential biases, advocating for human oversight to ensure ethical standards are upheld in peer review processes (Chustecki 2024).

In conclusion, while AI offers promising tools to enhance efficiency in reviewer selection and promote transparency in the peer review process, it is imperative to implement robust ethical guidelines and maintain human oversight to preserve the integrity and fairness of academic publishing.

Future directives

To address the ethical considerations in writing research papers using artificial intelligence, several future directives are essential:

- a) *Development of ethical guidelines*: establish comprehensive guidelines that delineate the ethical use of AI in research paper writing. These guidelines should cover transparency in AI involvement, proper attribution, and intellectual property rights.
- b) *Bias mitigation strategies*: implement strategies to identify and mitigate biases in AI algorithms. This involves using diverse and representative training datasets and regularly auditing AI systems for fairness.
- c) *AI transparency and accountability*: promote transparency in the use of AI by requiring researchers to disclose AI assistance in their papers. Accountability measures should also be instituted to ensure researchers are responsible for the outputs generated by AI tools.
- d) *Education and training*: provide education and training for researchers on the ethical use of AI. This includes understanding the limitations and potential biases of AI tools and the importance of ethical considerations in AI-assisted research.
- e) *Equitable access to AI tools*: ensure equitable access to AI-driven research tools across institutions, especially for those with fewer resources. This can help bridge the gap between resource-rich and resource-poor institutions, promoting inclusivity in academic research.

By adopting these directives, the academic community can effectively navigate the ethical landscape of AI in research paper writing, ensuring that AI technologies are used responsibly and equitably.

Conclusion

To summarize, the utilization of AI for writing research papers presents innovative opportunities as well as substantial ethical quandaries. Utilizing AI aid for writing research articles gives rise to complex and diverse ethical concerns, encompassing issues of authorship, intellectual property, data privacy, academic integrity, and bias. This work has provided a comprehensive understanding of the ethical considerations associated with the use of AI in academic research. It has also proposed a set of ethical standards for conducting research, based on an analysis of recent literature and case studies. To effectively respond to the evolving AI technology, future research should extensively investigate these challenges, establish comprehensive frameworks and policies to govern the ethical utilization of AI in academic writing, and promote continuous communication among researchers, ethicists, and politicians.

Abbreviations

AI Artificial intelligence
LLMs Large Language Models

Authors' contributions

NA, NSS and KST wrote main article, reviewed and corrected by AK and SKV.

Funding

None.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

¹All India Institute of Medical Sciences, Gorakhpur, India. ²Pt. Jawahar Lal Nehru Memorial Medical College, Raipur, India. ³Late Smt. Indira Gandhi Memorial Government Medical College, Kanker, India. ⁴University College of Medical Sciences, Delhi, India. ⁵Amrita School of Medicine, Amrita Vishwa Vidyapeetham, Faridabad, India.

Received: 29 January 2025 Accepted: 12 May 2025

Published online: 21 May 2025

References

- Amisha MP, Pathania M, Rathaur VK (2019) Overview of artificial intelligence in medicine. *J Family Med Prim Care* 8:2328–2331. https://doi.org/10.4103/jfmpc.jfmpc_440_19
- Bender E, McMillan-Major A, Shmitchell S, Gebru T (2021) On the Dangers of Stochastic Parrots: Can Language Models Be Too Big? FAccT '21: Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency 610–23. <https://doi.org/10.1145/3442188.3445922>
- Brown TB, Mann B, Ryder N, Subbiah M, Kaplan J, Dhariwal P, Neelakantan A, Shyam P, Sastry G, Askell A, Agarwal S, Herbert-Voss A, Krueger G, Henighan T, Child R, Ramesh A, Ziegler DM, Wu J, Winter C, Hesse C (2020) Language Models Are Few-Shot Learners. *arXiv.org* 4(33). <https://arxiv.org/abs/2005.14165>
- Bruce T, Amir Z, Allen BL, Alting BF, Amos M, Augusteyn J, Ballard GA, Behrendorff LM, Bell K, Bengsen AJ, Bennett A, Benshemesh JS, Bentley J, Blackmore CJ, Boscarino-Gaetano R, Bourke LA, Brewster R, Brook BW, Broughton C, Buettel JC, Luskin MS (2025) Large-scale and long-term wildlife research and monitoring using camera traps: a continental synthesis. *Biol Rev Camb Philos Soc* 100(2):530–555. <https://doi.org/10.1111/brev.13152>
- Bryson JJ, Theodorou A (2019) How society can maintain human-centric artificial intelligence. *IEEE Comput* 52(11):72–81. https://doi.org/10.1007/978-981-13-7725-9_16
- Cath C, Wachter S, Mittelstadt B, Taddeo M, Floridi L (2018) Artificial intelligence and the 'good society': The US, EU, and UK approach. *Sci Eng Ethics* 24(2):505–528. <https://doi.org/10.1007/s11948-017-9901-7>
- Causio FA, Angelis de L, Diedenhofen G, Talio A, Baglivo F, Workshop Participants (2024) Perspectives on AI use in medicine: views of the Italian Society of Artificial Intelligence in Medicine. *J Prev Med Hygiene*. 65(2):E285–E289. <https://doi.org/10.15167/2421-4248/jpmh2024.65.2.326>
- Chetwynd E (2024) Ethical Use of Artificial Intelligence for Scientific Writing: Current Trends. *J Hum Lact* 40(2):211–215. <https://doi.org/10.1177/08903344241235160>
- Chustecki M (2024) Benefits and Risks of AI in Health Care: Narrative Review. *Interact J Med Res* 13:e53616. <https://doi.org/10.2196/53616>
- Cross JL, Choma MA, Onofrey JA (2024) Bias in medical AI: implications for clinical decision-making. *PLOS Digital Health* 3(11):e0000651. <https://doi.org/10.1371/journal.pdig.0000651>
- Daneshjou R, Smith MP, Sun MD, Rotemberg V, Zou J (2021) Lack of transparency and potential bias in artificial intelligence data sets and algorithms: a scoping review. *JAMA Dermatol* 157(11):1362–1369. <https://doi.org/10.1001/jamadermatol.2021.3129>
- Daungsupawong H, Wiwanitkit V (2024) Large language model, AI and scientific research. *J Neurosurg Sci*. 68(4):500. <https://doi.org/10.23736/S0390-5616.24.06233-7>
- Fatima A, Shafique MA, Alam K, Fadlalla Ahmed TK, Mustafa MS (2024) ChatGPT in medicine: a cross-disciplinary systematic review of ChatGPT's (artificial intelligence) role in research, clinical practice, education, and patient interaction. *Medicine* 103(32):e39250. <https://doi.org/10.1097/MD.00000000000039250>
- Garcia MB (2024) Using AI tools in writing peer review reports: should academic journals embrace the use of ChatGPT? *Ann Biomed Eng* 52(2):139–140. <https://doi.org/10.1007/s10439-023-03299-7>
- George J (2024) Navigating the ethical maze of AI: balancing innovation, accountability, and human values. *Asian Horizons* 18(3):418–426 (<https://dvjournals.in/index.php/ah/article/view/4593/3755>)
- Greene D, Hoffmann A, Stark L (2019) Better, Nicer, Clearer, Fairer: A Critical Assessment of the Movement for Ethical Artificial Intelligence and Machine Learning. Proceedings of the 52nd Hawaii International Conference on System Sciences (HICSS) 2122–131. <https://scholarspace.manoa.hawaii.edu/server/api/core/bitstreams/849782a6-06bf-4ce8-9144-a93de4455d1c/content>
- Grieve J, Bartl S, Fuoli M, Grafmiller J, Huang W, Jawerbaum A, Murakami A, Perlman M, Roemling D, Winter B (2025) The sociolinguistic foundations of language modeling. *Front Artif Intel* 7:1472411. <https://doi.org/10.3389/frai.2024.1472411>
- Heidt A (2024) Intellectual property and data privacy: the hidden risks of AI. Nature. Epub ahead of print. PMID: 39232226. <https://doi.org/10.1038/d41586-024-02838-z>
- Hilliard W, Hobbs S, Jimenez R (2023) Exploring plagiarism. *Nursing* 53(9):35–39. <https://doi.org/10.1097/01.NURSE.0000942848.64158.dc>
- Hosseini M, Rasmussen LM, Resnik DB (2024) Using AI to write scholarly publications. *Account Res* 31(7):715–723. <https://doi.org/10.1080/08989621.2023.2168535>
- Hovy D, Spruit SL (2016) The social impact of natural language processing. Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics 2:591–8. <https://aclanthology.org/P16-2096.pdf>
- Inglada Galiana L, Corral Gudino L, Miramontes González P (2024) Ethics and artificial intelligence. *Rev Clin Esp* 224(3):178–186. <https://doi.org/10.1016/j.rceng.2024.02.003>
- Jairoun AA, El-Dahiyat F, ElRefae GA, Al-Hemyari SS, Shahwan M, Zyoud SH, Hammour KA, Babar ZU (2024) Detecting manuscripts written by generative AI and AI-assisted technologies in the field of pharmacy practice. *J Pharmaceut Policy Pract* 17(1):2303759. <https://doi.org/10.1080/20523211.2024.2303759>
- Jha N, Lee KS, Kim YJ (2022) Diagnosis of temporomandibular disorders using artificial intelligence technologies: a systematic review and meta-analysis. *PLoS ONE* 17(8):e0272715. <https://doi.org/10.1371/journal.pone.0272715>
- Juyal D, Thawani V, Thaledi S (2015) Rise of academic plagiarism in India: Reasons, solutions and resolution. *Lung India* 32(5):542–543. <https://doi.org/10.4103/0970-2113.164151>
- Kacena MA, Plotkin LI, Fehrenbacher JC (2024) The use of artificial intelligence in writing scientific review articles. *Curr Osteoporos Rep* 22(1):115–121. <https://doi.org/10.1007/s11914-023-00852-0>
- Kampa RK, Padhan DK, Karna N, Gouda J (2025) Identifying the factors influencing plagiarism in higher education: An evidence-based review of the literature. *Account Res* 32(2):83–98. <https://doi.org/10.1080/08989621.2024.2311212>
- King G, Pan J, Roberts ME (2017) How the Chinese government fabricates social media posts for strategic distraction, not engaged argument. *Am Polit Sci Rev*. 111(3):484–501. [cited 2025 February 03]. Available from: (<https://gking.harvard.edu/files/gking/files/50c.pdf>)
- Krishnan V (2024) Artificial intelligence in scientific peer review. *J World Fed Orthod* 13(2):55–56. <https://doi.org/10.1016/j.ejwf.2024.03.004>
- Kwon D (2024) AI is complicating plagiarism. How should scientists respond? Nature. Epub ahead of print. PMID: 39080398. <https://doi.org/10.1038/d41586-024-02371-z>
- Maroteau G, An JS, Murgier J, Hulet C, Ollivier M, Ferreira A (2023) Evaluation of the impact of large language learning models on articles submitted to Orthopaedics & Traumatology: Surgery & Research (OTSR): a significant increase in the use of artificial intelligence in 2023. *Orthopaed Traumatol, Surg Res* 109(8):103720. <https://doi.org/10.1016/j.otsr.2023.103720>
- Mittelstadt BD, Allo P, Taddeo M, Wachter S, Floridi L (2016) The ethics of algorithms: mapping the debate. *Big Data Soc* 3(2):2053951716679679. <https://doi.org/10.1177/2053951716679679>

- Mittermaier M, Raza MM, Kvedar JC (2023) Bias in AI-based models for medical applications: challenges and mitigation strategies. *NPJ Digital Medicine* 6(1):113. <https://doi.org/10.1038/s41746-023-00858-z>
- Mondal H, Mondal S (2023) ChatGPT in academic writing: Maximizing its benefits and minimizing the risks. *Indian J Ophthalmol* 71(12):3600–3606. https://doi.org/10.4103/IJO.IJO_718_23
- Nazer LH, Zatarah R, Waldrip S, Ke JXC, Moukheiber M, Khanna AK, Hicklen RS, Moukheiber L, Moukheiber D, Ma H, Mathur P (2023) Bias in artificial intelligence algorithms and recommendations for mitigation. *PLOS Digital Health* 2(6):e0000278. <https://doi.org/10.1371/journal.pdig.0000278>
- Nemitz P (2018) Constitutional democracy and technology in the age of artificial intelligence. *Philos Trans A, Math, Phys Eng Sci* 376(2133):20180089. <https://doi.org/10.1098/rsta.2018.0089>
- Noble S (2018) *Algorithms of Oppression: How Search Engines Reinforce Racism*. New York, New York University Press. <https://doi.org/10.18574/nyu/9781479833641.001.0001>
- Nussbaumer-Streit B, Klerings I, Dobrescu AI, Persad E, Stevens A, Garritty C, Kamel C, Affengruber L, King VJ, Gartlehner G (2020) Excluding non-English publications from evidence-syntheses did not change conclusions: a meta-epidemiological study. *J Clin Epidemiol* 118:42–54. <https://doi.org/10.1016/j.jclinepi.2019.10.011>
- O'Neil C (2016) *Weapons of math destruction: how big data increases inequality and threatens democracy*. New York, Crown Publishing Group 272. <https://dl.acm.org/doi/10.5555/3002861>
- Ong JCL, Chang SY, William W, Butte AJ, Shah NH, Chew LST, Liu N, Doshi-Velez F, Lu W, Savulescu J, Ting DSW (2024) Ethical and regulatory challenges of large language models in medicine. *Lancet Digit Health* 6(6):e428–e432. [https://doi.org/10.1016/S2589-7500\(24\)00061-X](https://doi.org/10.1016/S2589-7500(24)00061-X)
- Park YS (2024) White default: examining racialized biases behind AI-generated images. *Art Educ* 77(4):36–45. <https://doi.org/10.1080/00043125.2024.2330340>
- Pudasaini S, Miralles-Pechuán L, Lillis D, Llorens M (2024) Survey on plagiarism detection in large language models: the impact of chatgpt and gemini on academic integrity. [Preprint] <https://doi.org/10.48550/arXiv.2407.13105>
- Sandvig C, Hamilton K, Karahalios K, Langbort C (2014) Auditing algorithms: research methods for detecting discrimination on internet platforms. *Data Mining Ethics*. 24(2):434–51. [cited 2025 February 03]. Available from: (<https://websites.umich.edu/~csandvig/research/Auditing%20Algorithms%20-%20Sandvig%20-%20ICA%202014%20Data%20and%20Discrimination%20Preconference.pdf>)
- Stenseke J (2024) On the computational complexity of ethics: moral tractability for minds and machines. *Artif Intell Rev* 57(4):105. <https://doi.org/10.1007/s10462-024-10732-3>
- Tang A, Li KK, Kwok KO, Cao L, Luong S, Tam W (2024) The importance of transparency: Declaring the use of generative artificial intelligence (AI) in academic writing. *J Nurs Scholarsh* 56(2):314–318. <https://doi.org/10.1111/jnu.12938>
- Tennant JP (2018) The state of the art in peer review. *FEMS Microbiol Lett*. 365(19):fny204. <https://doi.org/10.1093/femsle/fny204>
- Wallach W, Franklin S, Allen C (2010) A conceptual and computational model of moral decision making in human and artificial agents. *Top Cogn Sci* 2(3):454–485. <https://doi.org/10.1111/j.1756-8765.2010.01095.x>
- Yousaf MN (2025) Practical Considerations and Ethical Implications of Using Artificial Intelligence in Writing Scientific Manuscripts. *ACG Case Rep J* 12(2):e01629. <https://doi.org/10.14309/crj.0000000000001629>

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.